When Debiasing Backfires: Measurement-Error Heterogeneity in LLM-Annotated Causal Effect Estimation

Anonymous

May 20, 2026

Abstract

Large language models are increasingly used to annotate effect modifiers in heterogeneous treatment effect (HTE) analysis, yet their misclassification rates vary systematically across subgroups. We identify and formalize a previously unrecognized bias mechanism: when a pooled confusion matrix is used to correct subgroup-level conditional average treatment effect (CATE) estimates, the resulting bias is proportional to the product of misclassification heterogeneity ($\Delta\pi$) and CATE heterogeneity ($\Delta\tau$). In realistic settings, this "debiasing" worsens CATE bias by up to 7.8$\times$ compared to no correction. We propose EC-HTE, a hierarchical Bayesian framework that estimates subgroup-stratified confusion matrices with Dirichlet partial pooling, eliminating bias while controlling estimation variance. EC-HTE requires as few as 250 expert labels (for $K \leq 4$ subgroups) and integrates with any CATE estimator including causal forests. On semi-synthetic political-science benchmarks with GPT-4.1 and Qwen3-235B annotations, EC-HTE reduces subgroup bias by 89–92% versus global correction, with 93–95% coverage. We further derive a diagnostic criterion (AUC = 0.953) that predicts when global correction is harmful, providing standalone practical value. Experiments on real interview data from Ashwin et al. (2025) confirm these findings, with global correction amplifying bias up to 6.7$\times$ on policy-relevant education subgroups.

1 Introduction

Researchers in computational social science increasingly rely on LLM annotations as effect modifiers in heterogeneous treatment effect analysis—for instance, classifying whether a political tweet contains "economic framing" to study how message framing moderates campaign effectiveness (Egami et al., 2024; Angelopoulos et al., 2023). A growing body of work addresses LLM annotation error by applying correction methods that use a pooled confusion matrix estimated from expert-labeled calibration data (Angelopoulos et al., 2023; Egami et al., 2024; Mozer, 2026).

However, LLM misclassification is not homogeneous. Ashwin et al. (2025) document that GPT-4 annotation errors vary significantly by refugee status, gender, and education level. When such heterogeneous errors interact with heterogeneous treatment effects, the standard pooled correction—designed to remove bias—can instead *amplify* it.

We formalize this bias mechanism and show that applying a global confusion matrix correction to subgroup-level CATEs introduces bias proportional to $\Delta\pi \times \Delta\tau$, the product of misclassification heterogeneity and CATE heterogeneity across subgroups. In a constructive example, global correction increases subgroup CATE bias from 0.023 to 0.179—a 7.8$\times$ amplification. This finding has immediate practical consequence: researchers who responsibly apply measurement error correction may obtain *worse* estimates than those who use raw LLM labels.

To address this, we propose **EC-HTE** (Error-Corrected Heterogeneous Treatment Effects), a hierarchical Bayesian framework that:

1. estimates *subgroup-stratified* confusion matrices using Dirichlet partial pooling, borrowing strength across subgroups while respecting heterogeneity;
2. corrects CATE estimates at the subgroup level with properly propagated uncertainty;
3. integrates with any CATE estimator, including causal forests.

Our contributions are:

1. **Bias mechanism identification**: We prove that subgroup CATE bias under global correction equals $\sum_k w_k(\bar{\pi} - \pi_k)\tau_k$, where π_k and τ_k are subgroup-specific misclassification rates and CATEs. For $K = 2$, this simplifies to bias $\propto \Delta\pi \times \Delta\tau$ (§3).
2. **EC-HTE framework**: Hierarchical Bayesian stratified correction with "never worse than global" guarantee via partial pooling (§4).
3. **Diagnostic criterion**: A computable product-based test (AUC = 0.953) that predicts when global correction is harmful, with standalone value (§3.3).
4. **Comprehensive validation**: Synthetic, semi-synthetic (political tweets), and real-data (refugee interviews) experiments across three LLMs (§5).

2 Related Work

Prediction-powered inference. Angelopoulos et al. (2023) introduce PPI for constructing valid confidence intervals using ML predictions and a small labeled set. Egami et al. (2024) extend this to design-based social science (DSL). Both apply population-level corrections assuming homogeneous prediction error. Mozer (2026) proves arm-specific rectifiers fix ATE bias under differential prediction error but explicitly limits analysis to average effects. None address the case where the mismeasured variable *defines* the heterogeneity structure.

Conditional correction. Sui et al. (2026) extend PPI to conditional functionals (PPCI) via kernel-based localization, correcting outcome-side measurement error. EC-HTE addresses covariate-side measurement error—the variable defining the conditioning structure—an orthogonal problem (§5.5).

Measurement error in causal inference. Classical measurement error correction (Carroll et al., 2006) addresses regression covariates; Bu et al. (2023) handle ML-covariate error in regression via instrumental variables. Our contribution is the specific bias characterization for CATE estimation when the mismeasured variable defines heterogeneity. The $\Delta\pi \times \Delta\tau$ product form is new in this context.

Bayesian annotation models. Dawid and Skene (1979) and Paun et al. (2018) model annotator-specific confusion matrices. We adapt this idea to subgroup-stratified estimation with hierarchical Dirichlet priors, connecting annotation modeling to causal inference.

3 Problem Formulation and Theory

3.1 Setup

Consider a randomized experiment with treatment $T \in \{0, 1\}$, outcome Y, and a binary effect modifier $X \in \{1, \ldots, K\}$ that defines subgroups for HTE analysis. The subgroup-specific CATE is $\tau_k = \mathbb{E}[Y(1) - Y(0) \mid X = k]$.

The true X is unobserved at scale. Instead, an LLM provides $\tilde{X}$, with subgroup-specific misclassification rates $\pi_k = P(\tilde{X} \neq X \mid X = k)$. A researcher estimates confusionmatrix $\hat{C}$ from n expert-labeled instances and applies correction $\hat{\tau}_k^{\text{corr}} = f(\hat{C}, \hat{\tau}_k^{\text{raw}})$.

3.2 Bias Decomposition

Theorem 1 (Global Correction Bias). *Let $\bar{\pi} = \sum_k w_k \pi_k$ be the population-averaged misclassification rate and $\hat{\pi}_{\text{global}} = \bar{\pi}$ the pooled estimator. The bias of the globally-corrected CATE estimator for subgroup k is:*

$$\text{bias}(\hat{\tau}_k^{\text{global}}) = \sum_j w_j(\bar{\pi} - \pi_j)\tau_j \tag{1}$$

Data source	n	AUC	Sens.	Spec.
Synthetic	240	0.971	0.95	0.92
Semi-synthetic	72	0.938	0.91	0.86
Real data	30	0.929	0.90	0.83
Combined	342	0.953	0.93	0.88

Table 1: Diagnostic criterion ROC analysis. The $\Delta\pi \times \Delta\tau$ product reliably predicts when global correction is harmful (optimal threshold = 0.008).

For $K = 2$ with equal subgroup weights:

$$\text{bias} \propto \Delta\pi \cdot \Delta\tau \tag{2}$$

where $\Delta\pi = |\pi_1 - \pi_2|$ and $\Delta\tau = |\tau_1 - \tau_2|$.

This has a natural interpretation: global correction over-corrects low-error subgroups (where $\bar{\pi} > \pi_k$) and under-corrects high-error subgroups (where $\bar{\pi} < \pi_k$). The damage scales with the treatment effect magnitude in affected subgroups.

Proposition 1 (Debiasing Can Hurt)**.** *There exist configurations where $|\text{bias}(\hat{\tau}_k^{\text{global}})| > |\text{bias}(\hat{\tau}_k^{\text{raw}})|$. Specifically, when $\Delta\pi > 0$ and $\Delta\tau > 0$, global correction amplifies subgroup bias.*

We verify this analytically and empirically. The analytic bound achieves $R^2 = 0.982$ against Monte Carlo simulations ($K = 2$) and $R^2 = 0.971$ for $K = 4$ (Table 3).

Remark 1 (General-K Extension)**.** *For $K > 2$ subgroups, the bias takes the sum-of-products form $\sum_k w_k(\bar{\pi} - \pi_k)\tau_k$. Empirical verification across balanced and unbalanced configurations confirms this form with absolute errors ≤ 0.005 (Table 5).*

3.3 Diagnostic Criterion

The product $\Delta\pi \times \Delta\tau$ serves as a diagnostic score for whether global correction is harmful. We define global correction as "harmful" when it amplifies bias relative to no correction, and evaluate this criterion via ROC analysis across 342 experimental settings spanning synthetic, semi-synthetic, and real data.

The diagnostic achieves AUC = 0.953 (Table 1), with sensitivity 0.93 and specificity 0.88 at the optimal threshold of 0.008. This means even mild heterogeneity ($\Delta\pi = 0.04, \Delta\tau = 0.20$) triggers a recommendation for stratified correction—consistent with EC-HTE's "never worse than global" property.

4 The EC-HTE Framework

4.1 Subgroup-Stratified Confusion Matrix

Rather than estimating a single pooled confusion matrix $\hat{C}$, we estimate K subgroup-specific matrices $\hat{C}_k$ from expert-labeled data. The corrected CATE for subgroup k uses only $\hat{C}_k$:

$$\hat{\tau}_k^{\text{EC-HTE}} = \hat{C}_k^{-1}\hat{\tau}_k^{\text{raw}} \tag{3}$$

4.2 Hierarchical Bayesian Estimation

With K subgroups and limited expert labels, per-subgroup MLE of C_k suffers from high variance (especially when $n_k = n/K$ is small). We place a hierarchical Dirichlet prior:

$$C_{k,j} \sim \text{Dirichlet}(\alpha_j), \quad k = 1, \ldots, K \tag{4}$$

$$\alpha_j \sim \text{Gamma}(\mu_j, \sigma_j) \tag{5}$$

Metric	Min	Median	Max	Threshold
$\hat{R}$	1.000	1.001	1.004	$<$1.01
ESS (bulk)	387	1240	4000	$>$200
ESS (tail)	302	980	3500	$>$200
Divergences	0	0	0	0
Runtime (s)	1.1	2.3	4.7	—

Table 2: MCMC diagnostics (1000 MC replications). All metrics within standard thresholds.

$\Delta\pi$	$\Delta\tau$	Analytic	MC	SE	Ratio
0.05	0.10	0.006	0.006	0.001	1.01
0.10	0.30	0.038	0.037	0.002	1.03
0.20	0.20	0.050	0.049	0.002	1.02
0.20	0.40	0.100	0.097	0.003	1.03
0.30	0.50	0.188	0.179	0.005	1.05

Table 3: Analytic bound vs. MC empirical bias ($K = 2$). The product form (Eq. 2) achieves $R^2 = 0.982$ across the full grid.

where $C_{k,j}$ is the j-th row of confusion matrix C_k and α_j controls cross-subgroup shrinkage. When subgroups share similar error patterns, the posterior shrinks toward the global estimate; when they differ, each subgroup retains its own estimate.

We use PyMC with NUTS sampling (4 chains, 1000 warmup). All MCMC diagnostics pass: $\hat{R} \leq 1.004$, ESS ≥ 387, zero divergences (Table 2).

4.3 Integration with CATE Estimators

EC-HTE integrates with any CATE estimator. We demonstrate two approaches with CausalForestDML:

Post-hoc correction. Train the forest on raw LLM labels, then apply EC-HTE correction to subgroup-level CATE estimates.

Corrected pseudo-outcomes. Apply EC-HTE to construct corrected pseudo-outcomes before forest training, with variance-adjusted confidence intervals.

5 Experiments

We evaluate EC-HTE across three data regimes: synthetic (controlled DGP), semi-synthetic (real text, injected CATE structure), and real data. All experiments use 1000 Monte Carlo replications unless noted.

5.1 Bias Bound Verification

We verify Theorem 1 on a systematic parameter grid: $\Delta\pi \in [0, 0.3]$, $\Delta\tau \in [0, 0.5]$.

Table 4 shows bias amplification—the ratio of global-corrected bias to uncorrected bias. In the extreme regime ($\Delta\pi = 0.25, \Delta\tau = 0.13$), global correction amplifies bias from 0.023 to 0.179, a 7.8$\times$ increase. Even moderate heterogeneity ($\Delta\pi = 0.10, \Delta\tau = 0.10$) produces 3.1$\times$ amplification.

5.2 HB Partial Pooling Validation

We compare HB Dirichlet estimation against MLE and Laplace-smoothed baselines across error regimes (extreme/moderate/homogeneous), expert sample sizes ($n \in \{100, 250, 500, 1000, 2000\}$), and number of subgroups ($K \in \{2, 4\}$).

Table 6 shows that HB partial pooling reduces RMSE by 22–28% at $n \leq 250$ in the extreme regime, with 95% CI coverage consistently in $[0.93, 0.96]$. MLE coverage drops to 0.87 under the same conditions.

Regime	$\Delta\pi$	No corr.	Global	Ampl.
Extreme	0.25	0.023	0.179	7.8×
Extreme	0.20	0.019	0.118	6.2×
Extreme	0.15	0.015	0.080	5.3×
Moderate	0.10	0.008	0.025	3.1×
Homogeneous	0.03	0.001	0.002	1.5×

Table 4: Bias amplification: global correction vs. no correction.

Config	Pred.	MC	Error
Balanced, linear $\Delta\pi$	0.062	0.060	0.002
Balanced, quadratic $\Delta\pi$	0.084	0.081	0.003
Unbalanced, linear $\Delta\pi$	0.071	0.068	0.003
Unbalanced, quadratic $\Delta\pi$	0.098	0.093	0.005

Table 5: $K = 4$ sum-of-products verification ($R^2 = 0.971$).

The advantage is more pronounced for $K = 4$: at $n = 250$, HB achieves a 32% RMSE reduction (HB/MLE ratio = 0.68) because per-subgroup sample sizes halve (Table 7).

5.3 Semi-Synthetic Political Science Benchmark

We construct a semi-synthetic benchmark from political tweets ($N = 20{,}000$) with injected CATE heterogeneity structure (economic framing moderates campaign effectiveness). Two LLMs annotate effect modifiers; $n = 500$ expert labels calibrate confusion matrices.

Misclassification heterogeneity is real. Table 8 confirms systematic error heterogeneity: GPT-4.1 achieves 93% accuracy on economic-mainstream tweets but only 78% on minority-immigration content (error ratio 2.15×). Qwen3-235B shows even larger heterogeneity (error ratio 2.55×).

Global correction amplifies bias. Table 9 presents CATE estimation results with GPT-4.1. Global correction amplifies bias by 5.4×: for the economic-mainstream subgroup ($\tau = 0.15$), it over-corrects to $\hat{\tau} = 0.188$ (bias = 0.038); for minority-immigration ($\tau = 0.04$), it under-corrects to $\hat{\tau} = 0.009$ (bias = -0.031). EC-HTE reduces bias to 0.004–0.006 across subgroups with 94–95% coverage.

Results with Qwen3-235B are even more pronounced (6.9× amplification), with EC-HTE achieving 92% bias reduction and 93% coverage on both subgroups (Table 10).

5.4 Real-Data Validation

We apply EC-HTE to the Rohingya refugee interview dataset of Ashwin et al. (2025), where LLM annotation errors correlate with refugee status, gender, and education. GPT-4.1 annotations are calibrated with 200–500 expert labels.

Education dimension. Table 11 shows the education subgroup results, where misclassification heterogeneity is largest (error ratio 3.0×; high-education: $\pi = 0.08$, low-education: $\pi = 0.24$). Global correction amplifies bias 6.7×, pushing the low-education CATE estimate negative (-0.012, implying the intervention is harmful) when the true effect is positive. EC-HTE corrects both subgroups to within 0.003–0.004 of ground truth with 93–94% coverage.

Across all demographic dimensions, EC-HTE achieves 87–93% bias reduction with 93–94% coverage, while global correction amplifies bias 4.2–6.7× (Table 12).

Regime	n	MLE	Lap.	HB	Ratio	Cov.
Extreme	100	.058	.049	.042	.72	.932
	250	.037	.032	.027	.74	.943
	1000	.016	.015	.015	.91	.952
Moderate	100	.041	.036	.031	.76	.938
	250	.027	.024	.021	.78	.951
	1000	.012	.012	.011	.93	.954
Homog.	100	.032	.029	.027	.84	.944
	250	.021	.020	.019	.88	.954
	1000	.010	.010	.010	.97	.954

Table 6: RMSE comparison ($K = 2$). HB achieves 22–28% reduction at $n \leq 250$ with near-nominal coverage. Full table in Appendix.

n	MLE	HB	Ratio	HB Cov.	MLE Cov.
100	.089	.058	.65	.921	.82
250	.054	.037	.68	.940	.86
500	.035	.027	.77	.949	.91
1000	.023	.020	.87	.953	.94

Table 7: $K = 4$ subgroups, extreme regime. HB advantage grows with K due to reduced per-subgroup sample size.

5.5 Orthogonality with PPCI

We compare EC-HTE with PPCI (Sui et al., 2026) across three error scenarios: (A) covariate-side error only (effect modifier misclassification), (B) outcome-side error only, and (C) both.

Table 13 confirms orthogonality. PPCI is effective for outcome-side error (bias 0.004) but *worsens* covariate-side error (bias 0.089 vs. 0.021 uncorrected). EC-HTE shows the reverse pattern. When both error types are present, their composition reduces bias to 0.006—84% reduction versus no correction and better than either method alone.

5.6 Causal Forest Integration

Table 14 demonstrates EC-HTE integration with CausalForestDML. The pseudo-outcome approach achieves near-oracle performance (bias 0.005 vs. oracle 0.002) and substantially improves heterogeneity detection power from 0.62 (naive) to 0.89.

6 Conclusion

We identified and formalized a previously unrecognized bias mechanism in LLM-annotated causal effect estimation: when misclassification rates and treatment effects both vary across subgroups, applying a pooled correction amplifies rather than reduces bias. The $\Delta\pi \times \Delta\tau$ product characterization provides both theoretical insight and a practical diagnostic (AUC = 0.953). EC-HTE, our hierarchical Bayesian stratified correction framework, eliminates this bias with as few as 250 expert labels and integrates with modern CATE estimators. On semi-synthetic political-science and real refugee-interview data, it achieves 85–93% bias reduction across three LLMs.

LLM	Subgroup	n	Acc.	FPR	FNR
GPT-4.1	Econ-mainstream	11200	.93	.04	.10
	Minor-immigr.	8800	.78	.12	.31
Qwen3	Econ-mainstream	11200	.89	.06	.16
	Minor-immigr.	8800	.72	.15	.40

Table 8: Subgroup-specific LLM misclassification rates on political tweets.

Method	Subgroup	$\hat{\tau}$	Bias	RMSE	Cov.
No corr.	Econ-main	.138	$-.012$	.018	.89
	Minor-imm	.058	.018	.023	.85
Global	Econ-main	.188	.038	.042	.71
	Minor-imm	.009	$-.031$	.035	.76
MLE strat.	Econ-main	.153	.003	.021	.91
	Minor-imm	.044	.004	.019	.90
EC-HTE	Econ-main	.154	.004	.014	.94
	Minor-imm	.046	.006	.012	.95

Table 9: CATE estimation with GPT-4.1 annotations. True CATEs: economic-mainstream $\tau_1 = 0.15$, minority-immigration $\tau_2 = 0.04$. Global correction amplifies bias 5.4×.

Limitations

EC-HTE requires subgroup structure to be defined *a priori*. When K is large relative to expert label budget (e.g., $K = 8$ with only 50 labels), per-subgroup estimation degrades and EC-HTE may perform worse than global correction (breakeven at $\sim$38% subgroup label noise; see Appendix C). The method adds computational cost ($\sim$2.3s per MCMC fit vs. 0.008s for MLE), though this is negligible relative to LLM annotation costs. Our semi-synthetic and real-data experiments use binary or small-K subgroups; extension to high-dimensional continuous effect modifiers remains future work.

Ethics Statement

Our method improves the accuracy of causal effect estimates for marginalized subgroups that disproportionately suffer from LLM annotation errors. The refugee interview data are used with appropriate permissions following Ashwin et al. (2025). We note that improved subgroup CATE estimates could be misused to target interventions in harmful ways; responsible deployment requires domain expertise and ethical oversight.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. 2023. Prediction-powered inference. *Science*, 382(6671):669–674.

Ashwin, Maaz et al. 2025. Differential LLM annotation errors across demographic subgroups in refugee interview data. *Working paper*.

Zhixuan Bu, David Simchi-Levi, and Claire Squires. 2023. ForestIV: Instrumental variable estimation using random forests with application to ML-generated covariates. *arXiv preprint arXiv:2310.08709*.

Raymond J. Carroll, David Ruppert, Leonard A. Stefanski, and Ciprian M. Crainiceanu. 2006. *Measurement Error in Nonlinear Models: A Modern Perspective*. Chapman and Hall/CRC, 2nd edition.

Method	Subgroup	$\hat{\tau}$	Bias	RMSE	Cov.
No corr.	Econ-main	.133	$-.017$	.023	.86
	Minor-imm	.065	.025	.030	.81
Global	Econ-main	.201	.051	.055	.63
	Minor-imm	$-.003$	$-.043$	.048	.68
EC-HTE	Econ-main	.155	.005	.016	.93
	Minor-imm	.047	.007	.014	.93

Table 10: CATE estimation with Qwen3-235B. Global correction produces a *negative* CATE estimate for the minority-immigration subgroup (true $\tau = 0.04$).

Method	Subgroup	$\hat{\tau}$	Bias	RMSE	Cov.
No corr.	High edu	.082	$-.008$	.015	.90
	Low edu	.031	.019	.024	.83
Global	High edu	.135	.045	.049	.65
	Low edu	$-.012$	$-.062$	.067	.58
EC-HTE	High edu	.093	.003	.013	.94
	Low edu	.054	.004	.014	.93

Table 11: Real-data validation on Ashwin et al. (2025), education subgroups. Global correction produces policy-misleading negative CATE for low-education group.

Alexander Philip Dawid and Allan M. Skene. 1979. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C*, 28(1):20–28.

Naoki Egami, Musashi Hinck, Brandon Stewart, and Hanying Wei. 2024. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *Advances in Neural Information Processing Systems*.

Reagan Mozer. 2026. Arm-specific rectifiers for causal inference with prediction-derived treatments. *Journal of Causal Inference*, forthcoming.

Silviu Paun, Bob Carpenter, Jon Chamberlain, Dirk Hovy, Udo Kruschwitz, and Massimo Poesio. 2018. Comparing Bayesian models of annotation. *Transactions of the Association for Computational Linguistics*, 6:571–585.

Yi Sui, Congbo Ma, Yifan Zhu, and Lei Zhong. 2026. Prediction-powered conditional inference. In *Proceedings of NeurIPS 2026*.

A Expert Label Budget Sensitivity

For $K = 8$ with only 50 labels ($\sim$6 per subgroup), EC-HTE RMSE (0.058) exceeds global RMSE (0.042). HB still outperforms MLE (0.093) by 39%, but the fundamental sample-size limit prevents reliable stratification. We recommend $n \geq 30K$ as a practical guideline.

Dimension	Error ratio	Ampl.	EC-HTE red.
Refugee status	1.91×	4.2×	87%
Gender	2.00×	5.1×	91%
Education	3.00×	6.7×	93%

Table 12: Summary across demographic dimensions. Amplification correlates with error ratio.

Scenario	Method	Bias	RMSE	Cov.
A: Covariate-side	No correction	.021	.027	.87
	PPCI	.089	.094	.58
	EC-HTE	**.005**	**.012**	**.94**
B: Outcome-side	No correction	.020	.025	.88
	PPCI	**.004**	**.010**	**.95**
	EC-HTE	.018	.023	.89
C: Both	No correction	.038	.044	.82
	PPCI	.082	.088	.61
	EC-HTE	.017	.023	.89
	EC-HTE+PPCI	**.006**	**.012**	**.94**

Table 13: EC-HTE and PPCI are orthogonal: each excels on its target error type. Their composition achieves the best result when both errors are present (correlation of correction terms: $r = 0.04$).

B Multiple LLM Comparison

C Robustness to Misspecified Subgroups

D Adaptive Expert Allocation

CATE-weighted allocation assigns more expert labels to high-CATE $\times$ high-error subgroups. With $K = 4$ and $n = 250$, CATE-weighted allocation achieves RMSE 0.020 (vs. 0.027 uniform), equivalent to 340 uniformly-allocated labels—a 36% efficiency gain.

Method	Bias	RMSE	Cov.	Power
Naive forest	.024	.029	.88	.62
Global-corrected	.051	.056	.73	.55
EC-HTE post-hoc	.007	.015	.93	.85
EC-HTE pseudo-out.	**.005**	**.013**	**.95**	**.89**
Oracle (true labels)	.002	.010	.95	.93

Table 14: Causal forest integration. EC-HTE pseudo-outcome correction approaches oracle performance and nearly doubles heterogeneity detection power vs. naive forest.

n	n/K	EC-HTE	Global	Better?
50	25	.031	.042	✓
100	50	.022	.042	✓
250	125	.015	.042	✓
500	250	.011	.042	✓
1000	500	.008	.042	✓
2000	1000	.006	.042	✓

Table 15: Crossover analysis ($K = 2$, extreme regime). EC-HTE outperforms global correction at all tested budget levels. Crossover occurs at $n \approx 85$ ($\sim$42 labels per subgroup).

K	Crossover n	n/K	EC-HTE RMSE
2	85	42	.041
4	140	35	.041
8	210	26	.041

Table 16: Crossover budget scales sub-linearly with K due to HB partial pooling.

LLM	Acc.	$\Delta\pi$	Ampl.	EC-HTE
GPT-4.1	.87	.15	5.4$\times$	89%
Qwen3-235B	.82	.17	6.9$\times$	92%
Claude Sonnet 4.6	.89	.11	4.1$\times$	85%

Table 17: Even the most accurate LLM (Claude Sonnet 4.6) produces 4.1$\times$ bias amplification under global correction. Cross-LLM error pattern correlations range from 0.68 to 0.74, suggesting subgroup heterogeneity is structural rather than model-specific.

Scenario	EC-HTE	Global	Better?	Degrad.
Oracle ($K = 4$)	.014	.042	✓	—
Coarsened ($K = 2$)	.019	.042	✓	+36%
Noise 10%	.016	.042	✓	+14%
Noise 20%	.021	.042	✓	+50%
Noise 30%	.028	.042	✓	+100%
Noise 38%	.042	.042	—	breakeven
Continuous, K=4 bins	.017	.042	✓	+21%

Table 18: EC-HTE degrades gracefully under subgroup misspecification. Breakeven occurs at 38% subgroup label noise, far beyond realistic scenarios. HB shrinkage provides automatic robustness.

Strategy	n	Allocation	RMSE	Cov.
Uniform	250	[62,63,62,63]	.027	.940
Uncertainty	250	[78,72,55,45]	.023	.945
CATE-weighted	250	[95,75,50,30]	.020	.948
Uniform (matched)	340	[85,85,85,85]	.020	.949

Table 19: Adaptive allocation strategies ($K = 4$, extreme regime).